

2022 IEEE International Conference on Blockchain (Blockchain) **Blockchain 2022**

Table of Contents

Message from the General Chairs	xiv
Message from the Steering Chairs	xvi
Message from the Program Chairs	xvii
Organizing Committee	xviii
Technical Program Committee	xix
Reviewers	

The 5th IEEE International Conference on Blockchain (Blockchain-2022)

Regular Papers

Permissionless Blockchain-Based Sybil-Resistant Self-Sovereign Identity Utilizing Attested Execution Secure Processors	1
<i>Koichi Moriyama (Institute of Information Security, Japan) and Akira Otsuka (Institute of Information Security, Japan)</i>	
Monitoring Provenance of Delegated Personal Data With Blockchain	11
<i>Chanyang Ju (Hanyang University, South Korea), Wenyi Tang (University of Notre Dame, USA), Changhao Chenli (University of Notre Dame, USA), Gwangwoon Lee (Hanyang University, South Korea), Jae Hong Seo (Hanyang University, South Korea), and Taeho Jung (University of Notre Dame, USA)</i>	
FPLotto: A Fair Blockchain-Based Lottery Scheme for Privacy Protection	21
<i>Yuechen Pan (Nankai University, China), Yiwen Zhao (Nankai University, China), Xiaoguang Liu (Nankai University, China), Gang Wang (Nankai University, China), and Ming Su (Nankai University, China)</i>	
Agent-Based Modelling of Bitcoin Consensus Without Block Rewards	29
<i>Benjamin Kraner (Blockchain & Distributed Ledger Technologies Group, University of Zürich, Switzerland), Sheng-Nan Li (Blockchain & Distributed Ledger Technologies Group, University of Zürich, Switzerland), Andreia Sofia Teixeira (LASIGE and Faculty of Sciences, University of Lisbo, Portugal), and Claudio J. Tessone (Blockchain & Distributed Ledger Technologies Group, University of Zurich, Switzerland)</i>	

SmartMixModel: Machine Learning-Based Vulnerability Detection of Solidity Smart Contracts .37...	
<i>Supriya Shakya (Indian Institute of Technology Patna, India), Arnab Mukherjee (RCC Institute of Information Technology, India), Raju Halder (Indian Institute of Technology Patna, India), Abyayananda Maiti (Indian Institute of Technology Patna, India), and Amrita Chaturvedi (Indian Institute of Technology (BHU), India)</i>	
The Philos Trust Algorithm: Preventing Exploitation of Distributed Trust .45.....	
<i>Pam Russell (University of Colorado Colorado Springs (UCCS)) and Philip N. Brown (University of Colorado Colorado Springs (UCCS))</i>	
Cryptocurrency Price Prediction with Multi-Task Multi-Step Sequence-to-Sequence Modeling .53....	
<i>Jesper Kristensen (Advanced Blockchain Research), Juan Pablo Madrigal-Cianci (Advanced Blockchain Research; SB-MATH-CSQI, École Polytechnique Fédérale de Lausanne, Switzerland), Giorgos Felekis (Advanced Blockchain Research), and Maria Liatsikou (No Affiliation)</i>	
Analysis of Polkadot: Architecture, Internals, and Contradictions .61.....	
<i>Hanaa Abbas (Hamad Bin Khalifa University, Qatar), Maurantonio Caprolu (Hamad Bin Khalifa University, Qatar), and Roberto Di Pietro (Hamad Bin Khalifa University, Qatar)</i>	
Blockchain-Based Federated Learning for Industrial Metaverses: Incentive Scheme with Optimal AoI .71.....	
<i>Jiawen Kang (Guangdong University of Technology (GDUT), China), Jiangtian Nie (Nanyang Technological University (NTU), Singapore), Jiang Xiao (Huazhong University of Science and Technology, China), Xianjun Deng (Huazhong University of Science and Technology, China), Siming Wang (Guangdong University of Technology, China), Zehui Xiong (Singapore University of Technology and Design, Singapore), Rong Yu (Guangdong University of Technology, China), and Dusit Niyato (Nanyang Technological University, Singapore)</i>	
Blockchain-Based Solutions for Education Credentialing System: Comparison and Implications for Future Development .79.....	
<i>Zoey Ziyi Li (Monash University, Australia), Joseph K. Liu (Monash University, Australia), Jiangshan Yu (Monash University, Australia), and Dragan Gasevic (Monash University, Australia)</i>	
Setchain: Improving Blockchain Scalability with Byzantine Distributed Sets and Barriers .87.....	
<i>Margarita Capretto (IMDEA Software Institute, Spain; Universidad Politécnica de Madrid, Spain), Martín Ceresa (IMDEA Software Institute, Spain), Antonio Fernández Anta (IMDEA Networks Institute, Spain), Antonio Russo (IMDEA Networks Institute, Spain; Universidad Carlos III de Madrid, Spain), and César Sánchez (IMDEA Software Institute, Spain)</i>	
Decentralized Health Data Distribution: A DLT-Based Architecture for Data Protection .97.....	
<i>Gioele Bigini (University of Urbino, Italy), Mirko Zichichi (Universidad Politécnica de Madrid, Spain), Emanuele Lattanzi (University of Urbino, Italy), Stefano Ferretti (University of Urbino, Italy), and Gabriele D'Angelo (University of Bologna, Italy)</i>	
Mosaic - A Blockchain Consensus Algorithm Based on Random Number Generation .105.....	
<i>Zhengzhong Sun (Technical University of Denmark, Denmark), Wei-Yang Chiu (Technical University of Denmark, Denmark), and Weizhi Meng (Technical University of Denmark, Denmark)</i>	

SegWit Extension and Improvement of the BlockSim Bitcoin Simulator .115.....	
<i>M. Basile (University of Pisa, Italy), Giovanni Nardini (Department of Ingegneria dell'Informazione, Italy; University of Pisa, Italy), Pericle Perazzo (University of Pisa, Italy), and Gianluca Dini (University of Pisa, Italy)</i>	
A Scalable Blockchain-Based Smart Contract Model for Decentralized Voltage Stability Using Sharding Technique .124.....	
<i>Kimia Honari (University of Alberta), Xiaotian Zhou (University of Alberta), Sara Rouhani (University of Manitoba), Scott Dick (University of Alberta), Hao Liang (University of Alberta), Yunwei Li (University of Alberta), and James Miller (University of Alberta)</i>	
A Distributed Clock Synchronization Protocol for Proof of Stake Blockchains .132.....	
<i>Yuya Miki (Tokyo Institute of Technology, Japan) and Kazuyuki Shudo (Tokyo Institute of Technology, Japan; Kyoto University, Japan)</i>	
Integrating Group Signatures in Complex Decentralized Marketplace Transactions for Improved Buyer Privacy .139.....	
<i>Sen Qiao (North Carolina State University, U.S.A), Varun Madathil (North Carolina State University, U.S.A), and Kemafor Anyanwu (North Carolina State University, U.S.A)</i>	
DeSAT: Towards Transparent and Decentralized University Counselling Process .149.....	
<i>Dhaval Thummar (National Institute of Technology Karnataka), Yerramaddu Jahnavi (National Institute of Technology Karnataka), Mudavath Prathyusha (National Institute of Technology Karnataka), Sayad Shahanaz (National Institute of Technology Karnataka), Bishakh Chandra Ghosh (Indian Institute of Technology Kharagpur), and Sourav Kanti Addya (National Institute of Technology Karnataka)</i>	
PrivChain: Provenance and Privacy Preservation in Blockchain Enabled Supply Chains .157.....	
<i>Sidra Malik (University of New South Wales), Volkan Dedeoglu (CSIRO), Salil S. Kanhere (University of New South Wales), and Raja Jurdak (Queensland University of Technology)</i>	
DevLeChain - An Open Blockchain Development Platform for Decentralized Applications .167.....	
<i>Wei-Yang Chiu (Technical University of Denmark, Denmark) and Weizhi Meng (Technical University of Denmark, Denmark)</i>	
Authenticated Multi-Version Index for Blockchain-Based Range Queries on Historical Data .177.....	
<i>Shlomi Linoy (University of New Brunswick, Canada), Suprio Ray (University of New Brunswick, Canada), and Natalia Stakhanova (University of Saskatchewan, Canada)</i>	

Short Papers

A New PoW Consensus of Blockchain Based on Legendre Sequence .187.....	
<i>Ye Yuan (Nankai University, China), Yiwen Zhao (Nankai University, China), Ming Su (Nankai University, China), Gang Wang (Nankai University, China), and Xiaoguang Liu (Nankai University, China)</i>	
Advancing Blockchain-Based Federated Learning through Verifiable Off-Chain Computations .194	
<i>Jonathan Heiss (TU Berlin, Germany), Elias Grünewald (TU Berlin, Germany), Stefan Tai (TU Berlin, Germany), Nikolas Haimel (TU Wien, Austria), and Stefan Schulte (TU Hamburg, Germany)</i>	

Block Interval Adjustment Based on Block Propagation Time in a Blockchain .202.....	<i>Masumi Arakawa (Tokyo Institute of Technology, Japan) and Kazuyuki Shudo (Kyoto University, Japan)</i>
Blockchain-Based Authenticated Stego-Channels: A Security Framework and Construction .208....	<i>Vikram Kanth (Naval Postgraduate School, USA) and Britta Hale (Naval Postgraduate School, USA)</i>
BSMFS: Blockchain Assisted Secure Multi-Keyword Fuzzy Search over Encrypted Data .216.....	<i>Partha Sarathi Chakraborty (Indian Institute of Technology Patna, India), Mangesh Shivaji Chandrawanshi (Indian Institute of Technology Patna, India), Puspesh Kumar (Indian Institute of Technology Patna, India), and Somanath Tripathy (Indian Institute of Technology Patna, India)</i>
CODE: Blockchain-Based Travel Rule Compliance System .222.....	<i>Chaehyeon Lee (POSTECH, Korea), Changhoon Kang (POSTECH, Korea), Wonseok Choi (POSTECH, Korea), Myunghun Cha (Coinone, Korea), Jongsoo Woo (POSTECH, Korea), and James Won-Ki Hong (POSTECH, Korea)</i>
Cost and Performance Analysis on Decentralized File Systems for Blockchain-Based Applications: State-Of-The-Art Report .230.....	<i>Aisyah Ismail (Aglive Lab, Australia), Mark Toohey (Aglive Lab, Australia), Young Choon Lee (Macquarie University, Australia), Zhongli Dong (The University of Sydney, Australia), and Albert Y. Zomaya (The University of Sydney, Australia)</i>
Decentralized Authorization using Hyperledger Fabric .238.....	<i>Muthukur Venkata Akhil Vasishta (Indian Institute of Technology Kharagpur, India), Balaji Palanisamy (University of Pittsburgh, USA), and Shamik Sural (Indian Institute of Technology Kharagpur, India)</i>
Fortuna: A Novel Staked Voting System for Distributed Pari-Mutuel Gaming .244.....	<i>Tucker Moore (Georgetown University, United States), Nathan Marshall (Georgetown University, United States), and Eric Burger (Georgetown University, United States)</i>
Graph Analysis of the Ethereum Blockchain Data: A Survey of Datasets, Methods, and Future Work .250.....	<i>Arijit Khan (Aalborg University, Denmark)</i>
On the Storage Overhead of Proof-of-Work Blockchains .258.....	<i>Alessandro Sforzin (NEC Laboratories Europe), Matteo Maso (AiSight), Claudio Soriente (NEC Laboratories Europe), and Ghassan Karame (Ruhr-University Bochum)</i>
Optimal Mining: Maximizing Bitcoin Miners' Revenues From Transaction Fees .266.....	<i>Mohsen Alambardar Meybodi (University of Isfahan, Iran), Amir Kafshdar Goharshady (Hong Kong University of Science and Technology, China), Mohammad Reza Hooshmandasl (University of Mohaghegh Ardabili, Iran), and Ali Shakiba (Vali-e-Asr University of Rafsanjan, Iran)</i>
PeloPartition: Improving Blockchain Resilience to Network Partitioning .274.....	<i>Juncheng Fang (University of California), Farzad Habibi (University of California), Kevin Bruhwiler (University of California), Fayzah Alshammari (University of California), Abhishek Singh (University of California), Yinan Zhou (University of California), and Faisal Nawab (University of California)</i>

Two-Stage Market-Based Task Allocation for Blockchain-Based Cyber-Physical Production Systems .282.....	
	<i>Larissa Kräemer (TU Dortmund University), Rico Ahlbäumer (TU Dortmund University), and Moritz Roidl (TU Dortmund University)</i>
λ - Constant Function Markets Generalizing and Mixing Automated Market Makers .290.....	
	<i>Giorgos Felekis (Advanced Blockchain Research) and Jesper Kristensen (Advanced Blockchain Research)</i>

Workshop Papers

BlockchainEvo Symposium 2022

A Data Science Pipeline for Algorithmic Trading: A Comparative Study of Applications for Finance and Cryptoeconomics .298.....	
	<i>Luyao Zhang (Duke Kunshan University, China), Tianyu Wu (Duke Kunshan University, China), Saad Lahrchi (Duke Kunshan University, China), Carlos-Gustavo Salas-Flores (Duke Kunshan University, China), and Jiayi Li (Duke Kunshan University, China)</i>
Analyzing Soft and Hard Partitions of Global-Scale Blockchain Systems .304.....	
	<i>Kevin Bruhwiler (University of California), Fayzah Alshammari (University of California), Farzad Habibi (University of California), Juncheng Fang (University of California), Yinan Zhou (University of California), Abhishek Singh (University of California), Ahmad Showail (Taibah University, Saudi Arabia), and Faisal Nawab (University of California)</i>
Block Pruning With UTXO Aggregation .312.....	
	<i>Taegy Song (Tokyo Institute of Technology, Japan) and Kazuyuki Shudo (Kyoto University, Japan)</i>
BLOCKVAC: A Universally Acceptable and Ideal Vaccination System on Blockchain .320.....	
	<i>Manika Sharma (International Institute of Information Technology, India), Kishore Kothapalli (International Institute of Information Technology, India), and Sujit Gujar (International Institute of Information Technology, India)</i>
CASC: Content Addressed Smart Contracts .326.....	
	<i>Markus Knecht (University of Zürich, Switzerland; Institute for Mobile and Distributed Systems IMVS, University of Applied Sciences Northwestern Switzerland, Switzerland) and Burkhard Stiller (University of Zürich, Switzerland)</i>
CredTrust: Credential Based Issuer Management for Trust in Self-Sovereign Identity .334.....	
	<i>Rahma Mukta (UNSW), Hye-young Paik (UNSW), Qinghua Lu (CSIRO), and Salil S. Kanhere (UNSW)</i>
Enhancing Scalability with Payment Requests Aggregation in Lightning Network .340.....	
	<i>Jungbeom Seo (Pohang University of Science and Technology, South Korea) and Jong Kim (Pohang University of Science and Technology, South Korea)</i>

Fairledger: a Fair Proof-of-Sequential-Work Based Lightweight Distributed Ledger for IoT Networks .348.....	Ronghua Xu (Binghamton University, USA) and Yu Chen (Binghamton University, USA)
Implementation of Digital Log-Book System for Lifts and Escalators Based on Blockchain Technology .356.....	Wai-Keung Yeung (Capax Technology Limited, China), Stephen M.K. Tong (Capax Technology Limited, China), Jerry Y.K. Wong (Electrical and Mechanical Services Department The Government of the Hong Kong Special Administrative Region of the People's Republic of China, China), Mentor Cheung (Electrical and Mechanical Services Department The Government of the Hong Kong Special Administrative Region of the People's Republic of China, China), Ashley K.Y. Cheung (Capax Technology Limited, China), Graham Lui (Electrical and Mechanical Services Department The Government of the Hong Kong Special Administrative Region of the People's Republic of China, China), and Kevin K.K. Cheng (Electrical and Mechanical Services Department The Government of the Hong Kong Special Administrative Region of the People's Republic of China, China)
LiftChain: A Scalable Multi-Stage NFT Transaction Protocol .362.....	Hari Kishore Chaparala (University of California, USA), Sai Vineeth Doddala (University of California, USA), Ahmad Showail (Taibah University, Saudi Arabia), Abhishek Singh (University of California, USA), Samaa Gazzaz (University of California, USA), and Faisal Nawab (University of California, USA)
Public Blockchain-Based Lightweight Anonymous Authentication Platform Using Zk-SNARKs for Low-Power IoT Devices .370.....	Jing Huey Khor (University of Southampton Malaysia, Malaysia), Michail Sidorov (Norwegian University of Science, Norway), Nathan Tze Min Ho (University of Southampton, United Kingdom), and Tze Hank Chia (University of Southampton, United Kingdom)
TAIRA-BSC - Trusting AI in Recruitment Applications Through Blockchain Smart Contracts .376..	Monirah Aleisa (University of Sussex, United Kingdom), Mona Alshahrani (University of Sussex, Al Imam Mohammad ibn Saud Islamic University, United Kingdom), Natalia Beloff (University of Sussex, United Kingdom), and Martin White (University of Sussex, United Kingdom)
Towards Achieving Confidentiality in Hyperledger Fabric .384.....	Benedikt Hofmann (Siemens AG, Germany), Prabhakaran Kasinathan (Siemens AG, Germany), and Martin Wimmer (Siemens AG, Germany)
You've Got a Friend in ME (Mobile Edge): Blockchain Processing with Cloud Node Backup .392....	Zane Karl (University of California, USA), Hayden Freedman (University of California, USA), Ahmad Showail (University of California, USA; Taibah University, Saudi Arabia), Abhishek Singh (University of California, USA), Samaa Gazzaz (University of California, USA), and Faisal Nawab (University of California, USA)

FBS Symposium 2022

- A Sealed-Bid Auction With Fund Binding: Preventing Maximum Bidding Price Leakage .398.....
Kota Chin (University of Tsukuba, National Institute of Information and Communications Technology, Japan), Keita Emura (National Institute of Information and Communications Technology, Japan), Kazumasa Omote (University of Tsukuba, National Institute of Information and Communications Technology, Japan), and Shingo Sato (Yokohama National University, Japan)
- Frax: A Fractional-Algorithmic Stablecoin Protocol .406.....
Sam Kazemian (Everipedia, Cayman Islands), Jason Huan (UCLA, USA), Jonathan Shomroni (Reichman University, Israel), and Kedar Iyer (Everipedia, USA)
- Privacy-Preserving Energy Trading Using Blockchain and Zero Knowledge Proof .412.....
Dongkun Hou (Xi'an Jiaotong-Liverpool University, China), Jie Zhang (Xi'an Jiaotong-Liverpool University, China), Sida Huang (Xi'an Jiaotong-Liverpool University, China), Zitian Peng (Xi'an Jiaotong-Liverpool University, China), Jieming Ma (Xi'an Jiaotong-Liverpool University, China), and Xiaohui Zhu (Xi'an Jiaotong-Liverpool University, China)
- Proposal of a Smart Contract-Based Security Token Management System .419.....
Shingo Fujimoto (University of Tsukuba, Japan; Fujitsu Limited, Japan) and Kazumasa Omote (University of Tsukuba, Japan)
- Time-Efficient Decentralized Exchange of Everlasting Options With Exotic Payoff Functions .427....
Juan P. Madrigal-Cianci (Advanced Blockchain Research) and Jesper Kristensen (Advanced Blockchain Research)
- Advanced Ledger: Supply Chain Management with Contribution Trails and Fair Reward Distribution .435.....
Takeshi Miyamae (Fujitsu Limited, Japan), Satoru Nishimaki (Fujitsu Limited, Japan), Makoto Nakamura (Fujitsu Limited, Japan), Takeru Fukuoka (Fujitsu Limited, Japan), and Masanobu Morinaga (Fujitsu Limited, Japan)

ISBM Symposium 2022

- Blockchain-Related Identity and Access Management Challenges: (De)Centralized Digital Identities Regulation .443.....
Rodolfo Mecozzi (EY Advisory S.p.A Milan, Italy), Giuseppe Perrone (EY Advisory S.p.A Rome, Italy), Dario Anelli (EY Advisory S.p.A, Italy), Nicola Saitto (EY Advisory S.p.A, Italy), Elia Paggi (EY Advisory S.p.A, Italy), and David Mancini (EY Advisory S.p.A, Italy)

BSS Workshop 2022

- Interwoven Hash of Vicious Circle Free Graph .449.....
Dominik Tomaszuk (University of Białystok, Institute of Computer Science, Poland), Szymon Głab (Łódź University of Technology, Poland), Filip Turoboś (Łódź University of Technology, Poland), Tomasz Pawlik (MakoLab S.A., Poland), Dominik Kuziński (MakoLab S.A., Poland), and Mirek Sopek (MakoLab S.A., Poland)

A Blockchain Implementation for Configurable Multi-Factor Challenge-Set Self-Sovereign Identity Authentication .455.....	
<i>Alex Nortä (Tallinn University, Estonia; Johannes Kepler University, Austria; Dymaxion OÜ, Estonia), Alexandr Kormiltsyn (Dymaxion OÜ, Estonia), Chibuzor Udokwu (University of Applied Sciences Upper Austria, Austria; Dymaxion OÜ, Estonia), Vimal Dwivedi (University of Tartu, Estonia; Dymaxion OÜ, Estonia), Sunday Aroh (Dymaxion OÜ, Estonia), and Ignas Nikolajev (Dymaxion OÜ, Estonia)</i>	
Cosmos Cash: Public Permissionless Approach Towards SSI and Use Cases .462.....	
<i>Andrea Giacobino (Ignite (formerly Tendermint)), David Grierson (Ignite (formerly Tendermint)), Har Preet Singh (Ignite (formerly Tendermint)), Patrick McHale (Ignite (formerly Tendermint)), and Simon Maggs (Ignite (formerly Tendermint))</i>	
Scalable Multi-Chain Coordination Via the Hierarchical Longest Chain Rule .468.....	
<i>Yanni Georgiades (University of Texas Austin), Karl Kreder (Austin), Jonathan Downing (Chief Architect Dominant Strategies), Alan Orwick (CEO Dominant Strategies), and Sriram Vishwanath (University of Texas Austin)</i>	
Data Integrity Verification in Network Slicing Using Oracles and Smart Contracts .476.....	
<i>João Paulo de Brito Gonçalves (Federal Institute of Espirito Santo (Ifes), Brazil), Gustavo Alochio (Federal University of Espirito Santo (Ufes), Brazil), Rodolfo da Silva Villaca (Federal University of Espirito Santo (Ufes), Brazil), and Roberta Lima Gomes (Federal University of Espirito Santo (Ufes), Brazil)</i>	

SPB Symposium 2022

A Blockchained Incentive Architecture for Federated Learning .482.....	
<i>João Paulo de Brito Gonçalves (Federal Institute of Espirito Santo (Ifes), Brazil) and Rodolfo da Silva Villaca (Federal University of Espirito Santo (Ufes), Brazil)</i>	
Security Analysis of DeFi: Vulnerabilities, Attacks and Advances .488.....	
<i>Wenkai Li (Hainan University, China), Jiuyang Bu (Hainan University, China), Xiaoqi Li (Hainan University, China), and Xianyi Chen (Hainan University, China)</i>	
A Survey on Safety Regulation Technology of Blockchain Application and Blockchain Ecology .494.....	
<i>Pengxu Shen (Hainan University, China), Suozai Li (China Electronics Corporation Hainan Joint Innovation Research Institute Co. Ltd, China), Ming Huang (China Electronics Corporation Hainan Joint Innovation Research Institute Co. Ltd, China), Haoyu Gao (Inner Mongolia University of Technology, China), Leixiao Li (Inner Mongolia University of Technology, China), Jun Li (Hainan University, Oxford-Hainan Blockchain Research Institute, China), and Hong Lei (Hainan University, SSC Holding Company Ltd, China)</i>	

Decentralized and Self-Sovereign Identity in the Era of Blockchain: A Survey .500.....	
<i>Yirui Bai (Hainan University, China), Hong Lei (Hainan University, China; SSC Holding Company Ltd., China), Suozai Li (China Electronics Corporation Hainan Joint Innovation Research Institute Co. Ltd, China), Haoyu Gao (Inner Mongolia University of Technology, China), Jun Li (Hainan University, China; Oxford-Hainan Blockchain Research Institute, China), and Leixiao Li (Inner Mongolia University of Technology, China)</i>	
IoT Data Storage on a Blockchain Using Smart Contracts and IPFS .508.....	
<i>João Paulo de Brito Gonçalves (Federal Institute of Espirito Santo, Brazil), Gabriel Spelta (Federal University of Espirito Santo (Ufes), Brazil), Rodolfo da Silva Villaça (Federal University of Espirito Santo (Ufes), Brazil), and Roberta Lima Gomes (Federal University of Espirito Santo (Ufes), Brazil)</i>	
Security Vulnerabilities and Countermeasures of Smart Contracts: A Survey .512.....	
<i>Epherem Merete Sifra (Xidian University)</i>	
A Minimal Disclosure Signature Authentication Scheme Based on Consortium Blockchain .516.....	
<i>Zelin Yang (Xidian University, China), Hui Ma (Xidian University, China), Mingchao Ai (Hangzhou Institute of Technology, Xidian University, China), Meijie Zhan (Guangzhou Institue of Technology, Xidian University, China), Gaofei Wu (Xidian University, China; Guilin University of Electronic Technology, China), and Yuqing Zhang (Hainan University, China; Xidian University, China; University of Chinese Academy of Sciences, China)</i>	
Author Index 523.	